Muqtadir Hussain

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Education

McMaster University September 2019 - April 2025
Bachelor of Engineering and Management (B.Eng.Mgt.), ***Engineering Physics*** *Hamilton, Ontario*

Skills

Programming & Development: MATLAB, Python, C/C++, Java (Spring Boot, Maven, Kafka), REST API, JavaScript, HTML/CSS, three.js, Embedded Systems, PID Control, Computer Vision (OpenCV, MediaPipe, YOLOv8), Git/Github

Modeling & Simulation: ANSYS Fluent, MCNP, Sentaurus TCAD (EDA), RSoft, Zemax, SPICE,

Instrumentation & Fabrication: CAD, 3D Printing, LIDAR/3D Metrology, FEA, GD&T, Microfabrication

Industrial: Technical Reporting, Root Cause Analysis, FMEA, Project Management, Stakeholder Liaison, R&D Support

Work Experience

Applied Precision Inc | Precision Metrology and Engineering Intern | Vaughan, ON Sept 2022 - Apr 2023

- Operated high-precision structured light and LIDAR scanners (Leica RTC 360, Zeiss Comet L3D, Faro) for reverse engineering and quality inspection across automotive, aerospace, and medical sectors.
- Processed and aligned point cloud data to generate watertight CAD models using Geomagic Design X and PolyWorks.
- Supported tooling inspections and component design workflows by preparing hybrid 3D/2D outputs for additive manufacturing and inspection reports.
- Contributed to technical research and marketing documentation while supporting logistics, sales, and documentation tasks.

Projects/Other Experience

Midas Core – J.P. Morgan Software Engineering Job Simulation August 2025

- Architected an event-driven ingestion pipeline with Spring Boot (Java 17) + Kafka, deserializing transaction events to decouple producers, absorb bursts, and support horizontal scaling (validated with embedded broker tests via Maven).
- Designed a relational domain model with JPA and an in-memory SQL store (H2 database); enforced ID/funds invariants and implemented transactional balance updates to ensure atomicity and consistency.
- Integrated an external Incentive REST service with timeouts and fail-open defaults; composed business logic to credit recipients (amount + incentive) while debiting senders (amount only), persisting a clear audit trail.
- Exposed a minimal read API (GET /balance) returning a typed DTO so users can see current balances; improved operability and testability via clean DI layering (listener → service → repository → controller) and self-contained H2 integration tests.

Autonomous Blackjack-Dealing System – Capstone Project Sep 2024 - Apr 2025

- Engineered and built a fully autonomous, gesture-controlled Blackjack dealer from concept to final demonstration, integrating a self-designed 3D-printed mechanical dispenser with an Arduino-controlled motor system that achieved an 85% single-card deal success rate
- The system's Python-based software used a MediaPipe neural network to classify 'Hit'/'Stand' hand gestures with 99.88% accuracy and a separate YOLOv8 model to recognize dealt cards, with all game states displayed through a custom Tkinter GUI.

CANDU-9 Simulator & Xenon–Transient Analysis Jan 2024 - Apr 2024

- Operated a CANDU-9 reactor simulator to perform full-cycle startup, shutdown, and transient operations including power maneuvers, poison override, and turbine trips.
- Simulated xenon transients using Python; predicted ^{135}Xe concentration and reactivity impacts during power changes and recovery scenarios; analyzed and validated core behavior under abnormal conditions by integrating feedback control mechanisms and reactivity limits.

Monte Carlo Reactor Simulation – Fuel Bundle Optimization Jan 2024 - Apr 2024

- Modeled neutron transport and energy deposition in CANDU fuel bundles using MCNP; compared natural uranium, enriched uranium, MOX, and thorium fuels to optimize power output, flux uniformity, and thermal performance for core stability and efficiency; Delivered detailed technical recommendations in a formal presentation

McMaster Nuclear Reactor – Nuclear Laboratory Series Sep 2023 - Dec 2023

- Performed neutron activation, radiographic imaging, and attenuation experiments using HPGe, NaI(Tl), and BF_3/He-3 detectors; analyzed decay kinetics, cross-sections, and neutron-material interactions to validate shielding models and non-destructive inspection techniques.

PID-Controlled Thermoelectric Oven/Cooler Jan 2022 - Apr 2022

- Designed & implemented closed-loop PID temperature control on a TI MSP430 Microcontroller (in C), interfacing an NTC thermistor via ADC and modulating a Peltier TEC with PWM; built a MATLAB GUI for UART-based live plotting, setpoint input, and data logging